

Central Arizona Regional Transportation Plan Proposal

May 2025 for Prof. Emily Finkel, USC Price PPD 633. Not in association with Central Arizona Governments (CAG). Edited down for writing sample.

1. OVERVIEW OF CAG

The Central Arizona Governments (CAG) was incorporated in 1975 and is one of six regional planning districts, or Councils of Government (COGs), which was established by Executive Order 70-2 and signed by the Governor of Arizona to provide effective regional planning services to Gila and Pinal counties. The CAG Region is comprised of Gila and Pinal counties, and includes the 17 incorporated communities of Apache Junction, Casa Grande, Coolidge, Eloy, Florence, Globe, Hayden, Kearny, Marana, Mammoth, Maricopa, Miami, Payson, Queen Creek, Star Valley, Superior and Winkelman. As described, the Ak-Chin Indian Community, Gila River Indian Community, San Carlos Apache Tribe, and White Mountain Apache Tribe are also Native community members of the region.

Growth and development in the neighboring metropolitan counties of Maricopa and Pima strongly influence travel patterns through Pinal County, as well as daily commutes to and from these counties for employment. In contrast, while Gila County has seen some minor urbanization, it remains largely rural, with a focus on activities such as mining, skiing, hiking, and camping. As a result, this RTP was developed with an awareness of the distinct characteristics of each county, while also recognizing that growth trends in Central Arizona and the influence of nearby metropolitan areas significantly impact Pinal County's population and contribute to recreational and tourism travel in the region's more rural areas.

2. GOALS, OBJECTIVES, KEY PERFORMANCE MEASURES

Vital to long-range transportation planning is the ability to assess progress toward regional priorities using goals, objectives, and performance measures to represent a comprehensive set of potential indicators that could inform future regional planning efforts. While this RTP does not cover all these measures in detail, they serve as a reference for tracking long-term progress and identifying priority areas for data collection, funding allocation, and infrastructure development.

2.1 Integrate Energy Conservation

The Arizona Department of Transportation (ADOT)'s Sustainable Transportation program encourages integration of renewable energy sources, such as solar-powered transit amenities, into transportation projects. This goal also harnesses the potential of the Federal Transit Administration (FTA) Low or No Emission Vehicle Program (49 USC § 5339(c)) which provides funding for the purchase or lease of low- and zero-emission transit vehicles, including electric, hydrogen, and hybrid buses. Arizona's Title 34 Energy Efficiency Standards also maintain a level of energy-efficient design in public infrastructure projects, including transportation facilities. Additionally, the RTP falls into alignment with the ADOT's Arizona State Freight Plan which advocates for reduction in truck-related emissions, supporting compliance with GHG standards.

- ❖ Objective 1: Incorporate energy-efficient design principles into transportation infrastructure, such as LED street lighting and solar-powered transit amenities.
 - Performance Measure 1: Percentage of transportation projects integrating LED lighting and solar infrastructure
- ❖ Objective 2: Encourage sustainable freight and logistics strategies to reduce fuel consumption and emissions from goods movement.
 - Performance Measure 1: Percentage of freight fleet using alternative fuels or fuel-efficient technologies
 - Performance Measure 2: Reduction in truck-related emissions in freight corridors

2.2 Provide Safe and Secure Facilities

The Arizona Strategic Highway Safety Plan (SHSP) aims to reduce fatalities and serious injuries on all public roads. Following key federal and state initiatives, this goal supports the overarching goal of the Arizona SHSP to create a safer place to live and work. This goal also addresses the critical aspect of security in transportation systems, which is a key component of federal transportation planning requirements. By focusing on incident reduction, improved surveillance, and real-time monitoring, CAG can not only address safety but also security concerns across all modes of transportation.

- ❖ Objective 1: Reduce injuries, fatalities and property damage for all modes of transportation.
 - Performance Measure 1: Total Crash Rate (per million vehicle miles traveled)
 - Performance Measure 2: Serious & Fatal Crash Rate (per million vehicle miles traveled)
- ❖ Objective 2: Minimize security risks on roadways and bikeways, and on public transportation facilities throughout the regional planning area.
 - Performance Measure 1: Number of reported incidents (crime, vandalism, etc.) on transit and bikeways.
 - Performance Measure 2: Increase in surveillance and lighting at transit stops and bike trails.

2.3 Encourage Transportation Electrification

- ❖ Objective 1: Expand electric vehicle (EV) charging infrastructure to support widespread EV adoption.
 - Performance Measure 1: Percent of highway corridors designated as EV-ready (with fast chargers every 50 miles)
 - Performance Measure 2: Increase in workplace and multi-family housing charging availability
- ❖ Objective 2: Increase adoption of electric transit and fleet vehicles.
 - Performance Measure 1 : Annual reduction in fleet vehicle emissions due to electrification
- ❖ Objective 3: Reduce barriers to EV adoption in disadvantaged and rural communities.
 - Performance Measure 1: Percent of EV chargers located in low-income or rural areas
 - Performance Measure 2: EV adoption growth rate in underserved communities compared to the regional average

3. CURRENT CONDITIONS:

Gila and Pinal Counties present distinct transportation planning and funding challenges due to their differing geographic and demographic characteristics. Gila County remains primarily rural, while Pinal County has experienced significant urbanization. This urban-rural distinction necessitates the careful allocation of transportation funding, as federal and state regulations restrict rural-designated funds from being used in urban areas and vice versa. As a result, determining how to allocate resources effectively while adhering to these funding restrictions remains a key challenge. The primary focus is on identifying opportunities to coordinate services and optimize resources to meet the diverse transportation needs of both rural and urban populations without violating programmatic funding regulations.

According to a 2024 Economic Conditions Report from the Arizona Office of Economic Opportunity, Arizona had one of the fastest growing populations in the United States from 2021-2022, ranking 5th among all U.S. states in numeric growth. Pinal County accounted for 7.7% of Arizona's total population growth from 2017 to 2022, constituting 6% of the state's total population in 2022. Notably, in 2022, 11.1% of individuals who moved to Pinal County from a different U.S. state were prime-working age adults (25 to 54 years). As one of the fastest-growing counties in Arizona, Pinal County has experienced significant population and economic expansion, resulting in heightened demand for roadway capacity, transit services, and multimodal infrastructure. The county's mix of urbanizing areas, industrial corridors, and planned developments necessitates strategic transportation planning to accommodate current and future mobility needs. In contrast, Gila County remains predominantly rural, with limited urban development and a transportation network that primarily serves recreational, agricultural, and resource-based industries. While transportation improvements in Gila County remain important, the scale and complexity of infrastructure needs in Pinal County require a more focused approach to ensure regional connectivity, economic competitiveness, and sustainable growth.

OnTheMap Census Data from 2022 was employed in the following study. 154,687 people live in the selection area but are employed outside, compared to only 31,775 employed within. Additionally, 35,901 live and work within the selection area. Pinal and Gila Counties have a heavy reliance on jobs outside the county (likely Maricopa County). Transportation planning must prioritize connectivity to external job markets as daily commutes on highways to employment centers in other counties may be substantial. Further findings show that the greatest number of workers from Pinal County head in the Northwest direction, and a substantial amount of people must commute long distances, with 19.7% traveling 25-50 miles and 17% traveling > 50 miles. Given Pinal County's major employment sectors (government, retail trade, healthcare) and Gila County's economy (mining, recreation, ranching, tourism), inter-county connections are vital in order to improve

arterial roads that connect communities within Pinal and Gila Counties, supporting residents commuting to local job sectors.

Occupied Housing Units:	Maricopa County	Pima County	Pinal County	Gila County
No Vehicle Available	6.20%	8.40%	4.40%	5.90%
1 Vehicle Available	36.10%	37.40%	37.90%	36.50%
2 Vehicles Available	38.80%	36.50%	37.90%	34.50%
3 Vehicles Available	13.40%	12.60%	13.70%	16.10%
4 Vehicles Available	4.10%	3.70%	4.30%	4.60%
5 or More Vehicles Available	1.40%	1.40%	1.90%	2.50%

Table 1. Vehicle Availability Across Households in Different Counties. Source: ACS 5-year Census Data. Created March 11, 2025.

Census information provided is available through the American Community Survey (ACS) 5-year estimates of 2018-2022. For comparative purposes, demographic information for Maricopa and Pima County (non-CAG) are also included where stated. Table 1 reveals that Maricopa and Pima counties have higher percentages of households with "No Vehicle Available" (6.20% and 8.40% respectively) compared to Pinal (4.40%) and Gila (5.90%). Higher vehicle ownership in rural areas may reflect the longer distances, limited public transit options, and potentially more dispersed activities in rural areas. Given this insight, regional planning should seek to maintain and enhance the existing roadway network to accommodate the dominant mode of personal vehicles. This would best be supplemented with targeted transit solutions that address specific needs within rural Central Arizona communities. Additional Census data revealed that across Maricopa, Pima, Pinal, and Gila Counties, the most popular mode of commute was via car, truck, or van with 76-79% of that percentage driving alone.

ADOT crash data is not yet publicly accessible; however, the Center for the Future of Arizona provides a tool that utilizes the same database as ADOT, offering an alternative means for analyzing traffic incidents across the state. In evaluating current transportation safety conditions in Central Arizona, crash data for Pinal and Gila counties is compared against Pima County. To provide insight on differences in traffic safety between urbanized and rural areas, Pima County was selected as a benchmark due to its inclusion of Tucson, the second-largest metropolitan area in the state after Phoenix.

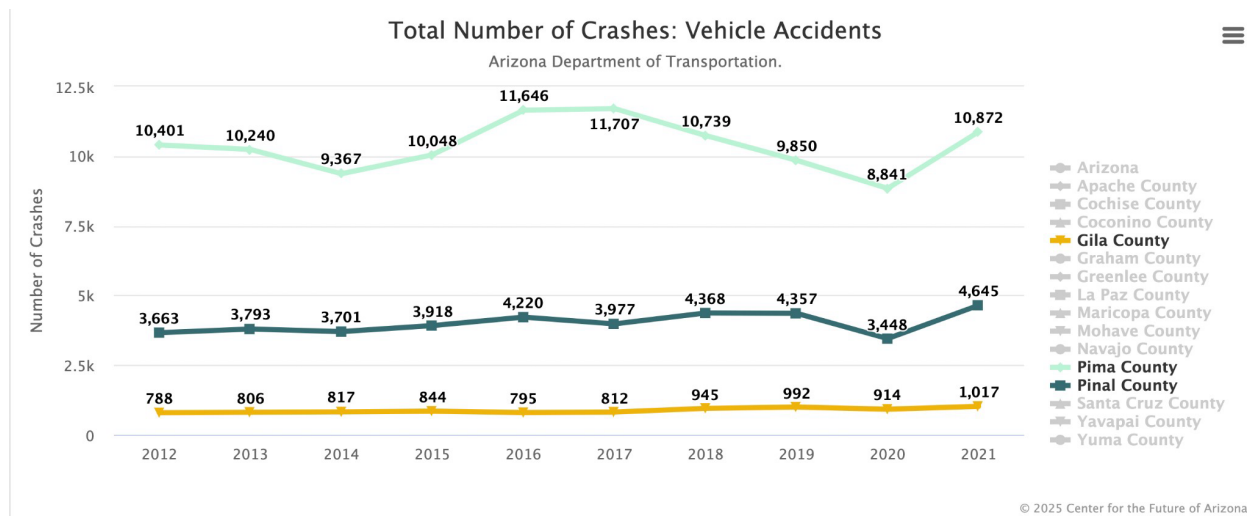


Figure A: Total Number of Vehicle Accident Collisions Among Arizona Counties from 2012-2022. Accessed March 11, 2025.

Figure A shows that Pima County consistently reports a significantly higher number of crashes involving fatalities, aligning with its higher population density and increased vehicle miles traveled. Pinal County, as a rapidly urbanizing region,

demonstrates a mid-range fatal crash rate, reflecting its mix of suburban, exurban, and rural land uses. In contrast, Gila County exhibits lower absolute numbers, which corresponds to its predominantly rural nature and lower traffic volumes.

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These findings highlight the need for region-specific transportation safety interventions, particularly in Pinal County, where ongoing development is likely to contribute to increased roadway risks.

4. LOOKING FORWARD TO AN ELECTRIC FUTURE:

As the transportation landscape evolves, electrification stands at the forefront of innovation and sustainability in Central Arizona. With the rapid growth of electric vehicle (EV) adoption statewide, the RTP recognizes the importance of preparing for an electrified future. This section explores the opportunities and challenges associated with electrification in Pinal and Gila counties, including equity considerations given lack of data, infrastructure needs, and environmental resilience.

4.1 Mapping Current Infrastructure

Arizona currently has 1,323 publicly accessible charging stations with 3,912 ports statewide. However, gaps remain along key corridors in Pinal and Gila counties. The NEVI program is addressing these gaps by funding fast-charging stations. Chargers are no more than 50 miles apart along designated Alternative Fuel Corridors, which currently include the interstate highways. However, EV adoption in Central Arizona remains relatively low compared to urban areas like Phoenix. To further reflect upon disparities in access to EV infrastructure, Native American communities, such as those served by the Three Rivers Tribes chapter of DRIVE Electric Arizona, are actively working to bridge this gap through education and advocacy for equitable EV access. Federal initiatives like NEVI and tribal-specific funding opportunities aim to address historic infrastructure inequities by ensuring that tribal lands are included in the EV transition. To support equity, resources such as grants for charging infrastructure deployment and workforce training will be critical.

3/11/2025

Interstate Highways of Arizona - AZINTRST — Other

Arizona Electric Vehicle Charging Stations

- 24 hours daily
- 24 hours daily, for customer use only
- 24 hours daily, for customer use only, see front desk for access
- 24 hours daily, for Tesla use only
- Other

County boundaries in Arizona

World_Hillshade

Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community, Esri, USGS

Figure B highlights a significant concentration of EV charging stations in and around Phoenix, particularly in Maricopa County. In contrast, Pinal County has a limited number of charging stations along I-10 and near Casa Grande, while Gila County has almost no charging infrastructure. To align with Arizona's climate action goals and the objectives set forth in the NEVI program, the RTP must prioritize equitable access to charging facilities across both urban and rural corridors.

The lack of adequate charging infrastructure along key highway corridors—including Interstate 10 (I-10), Interstate 8 (I-8), and U.S. 60—further inhibits regional connectivity and economic development. Addressing these deficiencies is essential to facilitating EV travel between major economic centers, supporting the state’s greenhouse gas (GHG) reduction targets, and ensuring compliance with alternative fuel corridor development plans. Furthermore, the absence of EV charging stations in rural and tourist-heavy areas, such as the Tonto National Forest and the San Carlos Reservation, limits the potential for economic growth in communities that could otherwise benefit from increased EV tourism.

Regional Policy Framework for Achieving EV Targets have been identified:

- **Prioritize the strategic deployment of EV charging infrastructure** along designated alternative fuel corridors, including I-10, I-8, and U.S. 60 to fill critical gaps in Casa Grande, Florence, Globe, and Payson.
- **Leverage state and federal funding opportunities**, including NEVI program grants and public-private partnerships, to accelerate the deployment of fast-charging stations at key mobility hubs.
- **Implement regulatory incentives and land-use policies** that mandate EV infrastructure integration in new commercial and residential developments.
- **Develop an EV infrastructure equity strategy** that ensures underserved and tribal communities receive equitable access to charging facilities.

4.2 Electrification of Freight

The electrification of freight rail and trucking presents both opportunities and challenges. While transitioning freight vehicles to electric power can significantly reduce emissions, it will require substantial upgrades to roadway infrastructure. The heavier battery sizes of electric trucks necessitate reinforced roadways to handle increased weight loads, particularly on major freight corridors like I-10 and I-8. Additionally, the deployment of high-capacity charging stations tailored for freight vehicles will be critical. These stations must support rapid charging times and accommodate large fleets, ensuring minimal disruption to logistics operations. Investments in reinforced pavements, expanded charging networks for heavy-duty vehicles, and intermodal hubs will be essential to support this shift. Additionally, strategic planning must align with federal freight programs such as the EPA SmartWay initiative to ensure sustainable goods movement across Central Arizona.

Electrifying freight transportation offers significant environmental benefits by reducing greenhouse gas emissions from one of Arizona's largest contributors to air pollution—transportation accounts for 40% of statewide emissions. This shift aligns with state and federal climate goals while also improving air quality in communities near major freight routes. The construction and maintenance of EV infrastructure, as well as advancements in battery technology and renewable energy integration, can position Central Arizona as a leader in sustainable logistics.

4.3 Climate Resilience and Extreme Conditions

Under challenging environmental conditions, Arizona must build a robust EV charging network capable of withstanding extreme heat and supporting widespread EV adoption. Modern EV chargers equipped with advanced cooling technologies will be necessary to maintain optimal performance during scorching temperatures. Further design choices such as shaded or climate-controlled charging stations protect both the equipment and vehicle batteries during charging. To proactively address these issues, charging infrastructure must be strategically located with placement along key evacuation routes or the most optimal siting to support emergency operations.

Central Arizona is looking to embrace electrification as a cornerstone of its transportation future.

By prioritizing equity and adapting to environmental realities, CAG can lead the region toward a cleaner, more sustainable transportation system that benefits all communities economically and environmentally.